

Database Management Systems — Reference Handbook

Database Management Systems (DBMS) · Unit 1 · Introduction & ER Model · 2024

Overview

These notes cover **Database Management Systems**. Design, query and manage relational databases — from ER modelling and SQL to normalization, transactions and recovery. Each section states the core idea, a worked intuition and the points examiners look for.

1. Database System Architecture

A DBMS separates the logical view of data from physical storage using a three-schema architecture: external, conceptual and internal levels.

Why it matters: Database System Architecture appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of database system architecture in one line.
- A small example showing database system architecture in practice.
- Common mistakes students make and how to avoid them.

2. Data Independence

Logical data independence lets the conceptual schema change without rewriting applications; physical data independence hides storage changes from the conceptual level.

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3. Entity-Relationship Model

ER diagrams model the real world as entities, attributes and relationships, and are the standard first step of database design.

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- Key definition of entity-relationship model in one line.
- A small example showing entity-relationship model in practice.
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4. Relational Algebra

Selection, projection, union, set difference, Cartesian product and rename form the procedural query language behind SQL.

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- Key definition of relational algebra in one line.
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5. Joins

Inner, natural, left, right and full outer joins combine tuples from relations based on a join predicate.

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- Key definition of joins in one line.
- A small example showing joins in practice.
- Common mistakes students make and how to avoid them.

6. Nested Queries

Subqueries with IN, EXISTS, ANY and ALL allow a query to use the result of another query, correlated or uncorrelated.

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- Key definition of nested queries in one line.
- A small example showing nested queries in practice.
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7. Functional Dependencies

A functional dependency $X \rightarrow Y$ states that tuples agreeing on X must agree on Y ; Armstrong's axioms derive all implied dependencies.

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- Key definition of functional dependencies in one line.
- A small example showing functional dependencies in practice.
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8. Closure of Attributes

The attribute closure X^+ is used to test candidate keys and to check whether a dependency is implied.

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- Key definition of closure of attributes in one line.
- A small example showing closure of attributes in practice.
- Common mistakes students make and how to avoid them.

9. First, Second and Third Normal Form

1NF requires atomic values, 2NF removes partial dependencies on a key and 3NF removes transitive dependencies.

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- Key definition of first, second and third normal form in one line.
- A small example showing first, second and third normal form in practice.
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10. ACID Properties

Atomicity, consistency, isolation and durability define correct transaction behaviour.

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- Key definition of acid properties in one line.
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11. Serializability

A schedule is conflict serializable if its precedence graph is acyclic, making it equivalent to some serial order.

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12. Two-Phase Locking

2PL has a growing and a shrinking phase; strict 2PL holds exclusive locks until commit to avoid cascading aborts.

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- A small example showing two-phase locking in practice.
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13. File Organization

Heap, sequential and hashed file organizations determine how records are placed on disk blocks.

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- Key definition of file organization in one line.
- A small example showing file organization in practice.
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14. B+ Tree Indexes

B+ trees keep data in sorted leaf nodes linked together, giving logarithmic search and efficient range scans.

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- Key definition of b+ tree indexes in one line.
- A small example showing b+ tree indexes in practice.
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15. Hash Indexes

Static and extendible hashing map search keys to buckets for constant-time equality lookups.

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- Key definition of hash indexes in one line.
- A small example showing hash indexes in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Database System Architecture · Data Independence · Entity-Relationship Model · Relational Algebra · Joins · Nested Queries · Functional Dependencies · Closure of Attributes · First, Second and Third Normal Form · ACID Properties · Serializability · Two-Phase Locking · File Organization · B+ Tree Indexes · Hash Indexes